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Tenure-track Associate Professor, Institute of Data and Information
Tsinghua Shenzhen International Graduate School, Tsinghua University, China

Education

Dec. 2013– Dec. 2016 **Ph.D.**, Computer Science, Paris-Saclay University (Advisor: [Prof. Frédéric Chazal](#))
Aug. 2011– Dec. 2013 **M.S.**, Computational Mathematics, Tsinghua University (Advisor: [Prof. Fengshan Bai](#))
Aug. 2007– Jun. 2011 **B.Sc.**, Mathematics and Applied Mathematics, Tsinghua University

Work Experience

Jul. 2025– present Tenure-track Associate Professor, [Tsinghua Shenzhen International Graduate School](#), Tsinghua University
Sep. 2020– Jun. 2025 Tenure-track Assistant Professor, [Tsinghua Shenzhen International Graduate School](#), Tsinghua University
Jan. 2017– Dec. 2019 Postdoctoral Researcher, [LIX](#), École Polytechnique (Advisor: [Prof. Maks Ovsjanikov](#))

Publications (in reverse chronological order)

 [Google Scholar](#)

† → Equal contribution * → Corresponding author

Journal Articles

- J18. Chen[†], Y., Zhang[†], G., Wang, M., Zhang, Y., Xie, J., Zhao, Z., **R.H***, Wu^{*}, J. & Dai^{*}, Q. Rapid robust high-fidelity 3D neuronal extraction from multiview calcium imaging datasets. *Nature Methods* (2026).
- J17. Chen[†], Z., Jiang[†], P., Sun[†], M. & **R.H***. NFR: Neural Feature-Guided Non-Rigid Shape Registration. *Computer Graphics Forum*, e70597 (2026).
- J16. Chen[†], Y., Tong^{†*}, C., Sun, J. Q., Shangguan, Y., Deng, X., Crowley, M., Wang, H., Ye, Y., Bao, H. & **R.H**. A Novel Soil Moisture Retrieval Method via Combining Radiative Transfer Model and Machine Learning. *Remote Sensing of Environment* (2026).
- J15. Yan[†], T., Guo[†], Y., Zhou[†], T., Shao, G., Li, S., **R.H***, Dai^{*}, Q. & Fang^{*}, L. A complete photonic integrated neuron for nonlinear all-optical computing. *Nature Computational Science* (2025).
- J14. Sun[†], M., Xu[†], C., Jiang[†], X., Liu, Y., Sun, B. & **R.H***. Beyond Talking – Generating Holistic 3D Human Dyadic Motion for Communication. *IJCV* (2025).

- J13. Xu[†], C., Sun[†], M., Cheng[†], Z.-Q., Wang, F., Liu, Y., Sun*, B., **R.H*** & Hauptmann, A. Combo: Co-speech holistic 3D human motion generation and efficient customizable adaptation in harmony. *IEEE TPAMI* (2025).
- J12. Zhang, T., Zheng*, B. & **R.H***. Adaptive high-resolution mapping of air pollution with a novel implicit 3D representation approach. *npj Climate and Atmospheric Science* (2025).
- J11. Lin[†], H., Gong[†], J., Wang[†], Y., Zhang[†], J., Bai, B., Zhang, Y., Wang, L., Wei, C., Cao, Y., Li, K., **R.H*** & Zhou*, G. AI system facilitates people with blindness and low vision in interpreting and experiencing unfamiliar environments. *npj Artificial Intelligence* (2025).
- J10. Shao[†], G., Zhou[†], T., Yan, T., Guo, Y., Zhao, Y., **R.H*** & Fang, L. Reliable, efficient, and scalable photonic inverse design empowered by physics-inspired deep learning. *Nanophotonics* (2025).
- J9. Yan[†], T., Zhou[†], T., Guo[†], Y., Zhao, Y., Shao, G., Wu, J., **R.H***, Dai*, Q. & Fang*, L. Nanowatt all-optical 3D perception for mobile robotics. *Science Advances* (2024).
- J8. Wan[†], J., Dong[†], Y., Yao, Y., Xiao, W., **R.H**, Xue, J.-H., Peng, R., Pei, H., Tian, X., Liao, R., He, H., Zeng, N., Li, C. & Ma*, H. Unsupervised Learning of Pixel Clustering in Mueller Matrix Images for Mapping Microstructural Features in Pathological Tissues. *Communications Engineering* (2023).
- J7. Wang, G., Zhang, J., Zhang, K., **R.H** & Fang*, L. GiganticNVS: Gigapixel Large-Scale Neural Rendering with Implicit Meta-Deformed Manifold. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023).
- J6. Wang[†], X., Chen[†], X., Jiang[†], P., Lin, H., Yuan, X., Ji, M., Guo, Y., **R.H** & Fang*, L. The Group Interaction Field for Learning and Explaining Pedestrian Anticipation. *Engineering* (2023).
- J5. **R.H**, Ren, J., Wonka, P. & Ovsjanikov, M. Consistent ZoomOut: Efficient Spectral Map Synchronization. *Computer Graphics Forum* (2020).
- J4. **R.H**, Achlioptas, P., Guibas, L. & Ovsjanikov, M. Limit Shape – A Tool for Understanding Shape Differences and Variability in 3D Model Collections. *Computer Graphics Forum* (2019).
- J3. **R.H** & Ovsjanikov, M. Adjoint Map Representation for Shape Analysis and Matching. *Computer Graphics Forum* (2017).
- J2. **R.H**, Chazal, F. & Ovsjanikov, M. On the Stability of Functional Maps and Shape Difference Operators. *Computer Graphics Forum* (2017).
- J1. Chazal, F., **R.H** & Sun, J. Gromov-Hausdorff Approximation of Filamentary Structures Using Reeb-Type Graphs. *Discrete & Computational Geometry* (2015).

Conference Papers

- C29. Chen[†], J., Kong[†], J., Chen, M., Zhang, B., Zhang, S., Zhao, H., **R.H***, Ma*, F. & Tian, Q. UniGarment: Topology-Guided Texture Normalization for Simulation-Ready Garment Digitization in *ACM Multimedia* (2026).
- C28. Wang[†], G., Lu[†], T., **R.H*** & Fang*, L. DANTE-W: Diffuse albedo neural texturing in the wild in *ECCV* (2026).

- C27. Chen[†], J., Zhang[†], B., Chen, M., Zhang, H., Zhang, S., Zhu, C., Zhao, H., **R.H***, Li, Z. & Wang*, Y. *One Video, One World: Turning Monocular Video into Physical 4D Scenes* in ECCV (2026).
- C26. Lu[†], T., Wang[†], G., **R.H*** & Fang*, L. *InceptionGS: Generative Bootstrapping for Large-Scale Gaussian Splatting under Unstructured View Sampling* in ECCV (2026).
- C25. Huang[†], G., Mao[†], J., Huang[†], F., Liu, F., Luo, X., Liang, Y., Lu, J., Wang, X., Liu, P., Fu*, R., **R.H*** & Huang*, S.-L. *Exposure bias can alleviate itself via directional and frequency rectification in flow matching* in ECCV (2026).
- C24. Jia^{†*}, Y., Zhang[†], H., Zhang[†], Z., Wu[†], J., Yu[†], M., Wang, Z., Jiang, D., Li, Z., Cao, C., Yu, Z., Yang, X., Ge, H., Zhang, Y., Zhang, J., Huang, Z., Liu, T., Chen, S., Wang, J., Xie, B., Yao, X., Deng, X., Wang, G., Zhang, J., Hao, L., Chen, Z., Chen, Y., Wang, A., Tian, H., Yan, Y., Cao, Z., Jiang, Y., Shao, H., Li, Y., Shi, L., Sui, W., Cui, H., Qin, Y., **R.H***, Chen, B., Han, L., Wang, T. & Zhou, G. *GS-Playground: A High-Throughput Photorealistic Simulator for Vision-Informed Robot Learning in Robotics: Science and Systems* (2026).
- C23. Chen, Z., Tao, J., Li, R., Hu, Y., Chen, R., Yang, Z., Yu, X., Jing, H., Zhang, M., Shao, S., Wang, B., Lu, Q. & **R.H***. *OmniVideo-R1: Reinforcing Audio-visual Reasoning with Query Intention and Modality Attention* in ICML (2026).
- C22. Sun[†], M., Zeng[†], C., Pei, J., Chen, J., Song, C., Wang, S., Chang, T., Huang, B., Zeng*, Z. & **R.H***. *Animator-Centric Skeleton Generation on Objects with Fine-Grained Details* in CVPR (2026).
- C21. Chen[†], Z., Zhang*, M., Yu, X., Luo, X., Sun, M., Pan, Z., An, X., Feng, Y., Pei, P., Cai, X. & **R. H***. *Think with 3D: Geometric Imagination Grounded Spatial Reasoning from Limited Views* in CVPR (2026).
- C20. Chen[†], J., Gao[†], K., Cui, Y., Sun, M., Chen, M., Wang, S., Long, X., Ma, F., Tian, Q., Zhao*, H. & **R.H***. *LottieGPT: Tokenizing Vector Animation for Autoregressive Generation* in CVPR (2026).
- C19. Chen, J., Chen, M., Xu, J., Li, X., Dong*, J., Sun, M., Jiang, P., Li, H., Yang, Y., Zhao*, H., Long, X.-X. & **R.H***. *Dance Together: Generating Interactive Multi-Person Video without Identity Drifting* in ICLR (2026).
- C18. Cai, Z., Yang, Y., Chang, X., Liang, S., Chen, R., Xiong*, F., Xu, M. & **R.H***. *Seeing Space and Motion: Enhancing Latent Actions with Geometric and Dynamic Awareness for Vision-Language-Action Models* in ICRA (2026).
- C17. Chen, Z., Zhao, R., Luo, C., Sun, M., Yu, X., Kang, Y. & **R.H***. *SIFThinker: Spatially-Aware Image Focus for Visual Reasoning* in AAAI (2026).
- C16. Cai[†], Z., Jiang[†], P., Mao, S., Cao*, H. & **R.H***. *Improving Multi-View Reconstruction via Texture-Guided Gaussian-Mesh Joint Optimization* in 3DV (2026).
- C15. Sun[†], M., Mao[†], S., Chen, K., Chen, Y., Lu, S., Wang, J., Dong*, J. & **R.H***. *ARMO: Autoregressive Rigging for Multi-Category Objects* in ICCV (2025).
- C14. Jia, Y., Wang, G., Dong, Y., Wu, J., Zeng, Y., Lin, H., Wang, Z., Ge, H., Gu, W., Ding, K., Yan, Z., Cheng, Y., Li, Y., Wang, Z., Li, C., Sui, W., Shi, L., Tian, G., **R.H*** & Zhou*, G. *DISCOVERSE: Efficient Robot Simulation in Complex High-Fidelity Environments* in IROS (2025).

- C13. Sun[†], M., Chen[†], J., Dong^{*}, J., Chen, Y., Jiang, X., Mao, S., Jiang, P., Wang, J., Dai, B. & **R.H***. *DRiVE: Diffusion-based Rigging Empowers Generation of Versatile and Expressive Characters* in CVPR (2025).
- C12. Chen[†], Z., Jiang[†], P. & **R.H***. *DV-Matcher: Deformation-based Non-Rigid Point Cloud Matching Guided by Pre-trained Visual Features* in CVPR (2025).
- C11. Sun[†], M., Chen[†], G., Jiang[†], P., Mao, S., Chen, Y. & **R.H***. *SRIF: Semantic Shape Registration Empowered by Diffusion-based Image Morphing and Flow Estimation* in SIGGRAPH Asia (2024).
- C10. Mao, S., Sun, M. & **R.H***. *Enhancing Human Optical Flow via 3D Spectral Prior* in Pacific Graphics (2024).
- C9. Wang, G., Zhang, J., Wang, F., **R.H*** & Fang^{*}, L. *XScale-NVS: Cross-Scale Novel View Synthesis with Hash Featurized Manifold* in CVPR (2024).
- C8. Ying, H., Yin, Y., Zhang, J., Wang, F., Yu, T., **R.H** & Fang^{*}, L. *OmniSeg3D: Omniversal 3D Segmentation via Hierarchical Contrastive Learning* in CVPR (2024).
- C7. Sun[†], M., Mao[†], S., Jiang, P., Ovsjanikov, M. & **R.H***. *Spatially and Spectrally Consistent Deep Functional Maps* in ICCV (2023).
- C6. Jiang[†], P., Sun[†], M. & **R.H***. *Non-Rigid Shape Registration via Deep Functional Maps Prior* in NeurIPS (2023).
- C5. Jiang, P., Sun, M. & **R.H***. *Neural Intrinsic Embedding for Non-rigid Point Cloud Matching* in CVPR (2023).
- C4. Lin, H., Chen, Z., Zhang, J., Bai, B., Wang, Y., **R.H** & Fang^{*}, L. *RealGraph: A Multiview Dataset for 4D Real-World Context Graph Generation* in ICCV (2023).
- C3. Zhang, J., Tang, R., Cao, Z., Xiao, J., **R.H*** & Fang^{*}, L. *ElasticMVS: Learning elastic part representation for self-supervised multi-view stereopsis* in NeurIPS (2022).
- C2. Ying[†], H., Zhang[†], J., Chen, Y., Cao, Z., Xiao, J., **R.H*** & Fang^{*}, L. *ParseMVS: Learning Primitive-aware Surface Representations for Sparse Multi-view Stereopsis* in ACM MM (2022).
- C1. **R.H[†]**, Rakotosaona[†], M.-J., Achlioptas, P., Guibas, L. & Ovsjanikov, M. *OperatorNet: Recovering 3D Shapes From Difference Operators* in ICCV (2019).

Research Support

2024–2028	High-precision Computational Photography for All-time and All-region Lunar Exploration, NSFC Key Program, Subproject Leader, CNY 0.70M allocated.
2023–2025	Noninvasive Dynamic Biomechanics Study of Organoids Based on Mueller Microscopy Imaging, SIGS Cross-disciplinary Fund, PI, CNY 0.125M allocated.
2022–2025	3D Reconstruction of Dynamic Scenes Based on Spectral Methods: Theory and Technique, NSFC General Program, PI, CNY 0.56M.
2022–2023	Depth Image Enhancement and 3D Reconstruction, Berxel Photonics Co., Ltd., PI, CNY 0.45M.

2022–2024 Shenzhen Key Laboratory for Next-Generation Interactive Media Technology Innovation, Shenzhen Science, Technology and Innovation Commission, participant, CNY 0.16M allocated.

Granted Patents

- F. Zhang, J. Gao, **R. Huang**, and M. Xin. “Method for Constructing a Fluorescence Microscopy Signal Detection Model and Application Thereof.” Chinese invention patent ZL 2024 1 1736891.7; authorization announcement CN 119693942 B; granted Oct. 28, 2025.
- **R. Huang**, J. Wang, and C. Guo. “Real-time Binocular RGB-D Stitching Method Based on 3D Geometric Properties.” Chinese invention patent ZL 2024 1 1474575.7; authorization announcement CN 119444586 B; granted Sep. 23, 2025.
- **R. Huang**, T. Zhang, and B. Zheng. “Method for Mapping 3D Air-Pollutant Concentrations Based on a High-field Signed Distance Function.” Chinese invention patent ZL 2024 1 0348218.X; authorization announcement CN 118332059 B; granted Sep. 9, 2025.
- **R. Huang**, M. Sun, S. Mao, P. Jiang, and F. Su. “Non-rigid Shape Registration Method and Storage Medium.” Chinese invention patent ZL 2023 1 0710920.1; authorization announcement CN 116703991 B; granted Feb. 17, 2026.
- **R. Huang**, P. Jiang, M. Sun, and B. Li. “Non-rigid Point Cloud Registration Method and Storage Medium.” Chinese invention patent ZL 2023 1 0691274.9; authorization announcement CN 116681744 B; granted Nov. 25, 2025.

Professional Service

- Local Arrangements Co-Chair for IVMSP 2026.
- Co-organizer of the inaugural CVPR 2025 workshop, “Enforcing Geometric, Physical, Topological, and Functional Inductive Biases in 3D Generation.”
- Member of the CAAI Expert Committee on Metaverse Technologies (2022–present); member of IEEE and ACM.
- Reviewer for journals including IJCV, ACM TOG, IEEE TVCG, IEEE TIP, IEEE TMM, and Graphical Models, and for conferences including CVPR, ICCV, ECCV, NeurIPS, ICLR, SIGGRAPH, SIGGRAPH Asia, Eurographics, and Pacific Graphics (2020–present).